Single Agent Systems & Agent Pipelines - Quiz

Instructions:
Answer the following questions. These questions are designed to test your understanding of agent systems, tools, and evaluation.

1. Explain the concept of a stateful directed graph in agent pipelines. How does it differ from a simple linear pipeline?

⇨ A stateful directed graph is a workflow where each step can remember information from previous steps. It can also choose different paths or repeat steps. A simple linear pipeline follows one fixed order from start to finish.

2. Describe the role of nodes and edges in an agent workflow. Give an example of each.

⇨ Nodes are the tasks done by the agent. Edges are the connections between the tasks. Example: A Calculator Tool is a node. The connection from the question to the Calculator Tool is an edge.

3. What is conditional routing in an agent system? Design a simple rule-based routing logic for three different query types.

⇨ Conditional routing means choosing a different action based on the type of question.
Example:
- If the question says "calculate" → Calculator Tool
- If it says "keywords" → Keyword Tool
- Otherwise → General Response

4. Why are cycles (loops) important in agent pipelines? Provide a use case where a retry loop is necessary.

⇨ Loops allow the agent to try a task again when it fails. Example: If an API request fails because of a network problem, the agent can try the request again instead of stopping.

5. Explain how a single-agent system can simulate multi-agent behavior internally.

⇨ A single agent can divide its work into different roles. It can first understand the question, then choose a tool, and finally give the answer. So, one agent can do work similar to multiple agents.

6. What are JSON schema tools? How do they help in structuring tool inputs and outputs?

⇨ JSON schema tools define what information a tool needs and what format it should have. They make sure the input is correct and the output follows a clear format. This helps reduce errors.

7. Compare sequential tool calls and parallel tool calls. When would you prefer one over the other?

⇨ Sequential tool calls happen one after another. They are useful when one task depends on the previous task. Parallel tool calls happen at the same time. They are useful when the tasks are independent and do not depend on each other.

8. How would you implement error handling in a tool-using agent? Provide at least two strategies.

⇨ Two simple ways are :
1. Use try-except to catch errors.
2. Use a retry system to try the failed task again.
We can also save errors in logs to find and fix problems later.

9. What is trajectory evaluation in agent systems? Why is it important beyond just checking final output?

⇨ Trajectory evaluation means checking all the steps the agent took, not only the final answer.
It helps us find mistakes in the agent's decisions, tool calls, and intermediate steps.

10. Define task completion rate and cost metrics. How would you measure and optimize them in a real-world system?

⇨ Task completion rate tells us how many tasks the agent completed successfully. For example, completing 90 out of 100 tasks means a 90% completion rate. Cost metrics measure things like API calls, time, or money used. We can improve the system by reducing unnecessary tool calls and using efficient methods.